QUANTUM MECHANICS FROM THE TORSION MEDIUM

Author: Robert Jobson | Date: 2026-08-21

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FRAMING NOTE (added 2026-08-27, v4; corrected 2026-09-02):

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The photon descriptions in this document are quantitatively correct.
The correct qualitative picture, developed in subsequent work, is that
a photon is a localized T_1g wave packet (corpuscle, Newton's term) --
not an extended sinusoidal wave. What propagates as "light" is the T_1g
TRANSVERSE vector wave in the medium (not a compression/pressure wave --
see doc_torsion Section 3.1 and doc_jobson_cell Section 7.1: the v_p
compression branch is carried by A_g, a separate scalar mode); the photon
corpuscle is the quantized, localized form of that transverse excitation.
Wave-particle duality in standard QM arises from conflating the medium
wave with the localized packet.
Note: T_1g waves may also be generated by macroscopic charge oscillations
(classical EM) without individual corpuscle events -- whether all T_1g
propagation requires a corpuscle source is not yet derived. [OPEN] This
mirrors a real, unresolved distinction in mainstream quantum optics, not
a torsionverse-specific gap: even canonical "single-photon" double-slit
demonstrations typically use an attenuated ("faint") laser for the
photon CORPUSCLE crossing the slits -- a Poissonian source with
g^2(0)=1, no antibunching, meaning "attenuated" only lowers the
average photon (corpuscle) number, it does not guarantee one discrete
emission event per trial.
True single-photon sources (heralded SPDC, quantum dots, NV centers)
reliably emit one photon corpuscle per trigger, shown by genuine
antibunching (g^2(0) -> 0), and are experimentally distinguishable
from faint-laser sources by this criterion. The
quantum eraser experiment cited below (Kim et al. 2000) is itself an
example: the photon CORPUSCLE crossing its double slit comes from an
attenuated argon laser, not a heralded single-photon source -- only
the later SPDC-generated entangled corpuscle pair is unambiguously
non-classical.
An electron is a corpuscle locked into a stable (1,2) Hopf winding;
the winding gives structure to its medium wave. A photon corpuscle carries
no persistent winding (theta=0). Both are corpuscles; the winding is the
only difference. All mathematical results in this document are unchanged.
Note: (1,2) is the shared winding class of essentially all matter
particles (protons included, Section 1.2) -- scale/zone/N_J, not the
(p,q) pair, is what distinguishes electron from proton.
The quantized, corpuscular nature of light -- established by the photoelectric
effect -- is consistent with localized T_1g wave packets. Newton lacked the
medium to explain the accompanying wave behavior.
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ABSTRACT

The Schrödinger equation is not a postulate. It is the non-relativistic limit of the Klein-Gordon equation that governs Hopf winding modes in the torsion medium. The derivation chain has zero free parameters beyond m_p (the framework's one external input, as in the companion papers):

1. Torsion medium: K = 1/eps_0, rho = mu_0, c = sqrt(K/rho) [PROVEN]
2. Free massless wave: omega = c*k (mode-neutral template) [QM1]
3. Hopf winding rest energy = m_p*c^2 -> Compton frequency omega_C = m_p*c^2/hbar
   [QM2; the Compton wavelength IS lambda_p = hbar_c/m_p = Zone 1 boundary]
4. Winding dispersion: omega^2 = c^2*k^2 + omega_C^2 [Klein-Gordon, QM3]
5. Non-relativistic limit (v << c):
     omega_KG -> omega_C + hbar*k^2/(2*m_p)
   Remove rest-mass oscillation phi = psi * exp(-i*m_p*c^2*t/hbar):
     i*hbar * dpsi/dt = -(hbar^2/2*m_p) * nabla^2 * psi [SCHRODINGER, QM5]

```
6. Born rule: P(x) = |psi(x)|^2 from medium wave energy density  [QM6]
   (a field-level/ensemble argument; Section 4.1 is explicit about what
   remains open at the level of an individual detection event)
```

The Schrödinger equation is quantum mechanics. It is now derived from the same medium (the torsion medium) that gives alpha, G, the Higgs mass, and quantum entanglement. No additional postulates or free parameters beyond m_p for steps 1-5 (the Schrödinger equation itself); step 6's field-level Born-rule argument leaves one open question, addressed in Section 4.1.

NEW GROUND: no existing interpretation of quantum mechanics -- Copenhagen, many-worlds, de Broglie-Bohm, GRW, QBism, consistent histories -- derives the Schrödinger equation, the Born rule, or the Tsirelson bound from an underlying physical medium with independently fixed mechanical properties. Every existing approach postulates the Hilbert-space/probability structure as an axiom. This paper derives it instead, as a mechanical consequence of the same medium that gives alpha, G, and the Higgs mass in the companion papers. (The Born rule claim here is at the field/ensemble level; see Section 4.1 for the open individual-event question, not resolved by this paper either.)

Consequences:

```
- Double-slit: the corpuscle takes one definite slit; its Zone 3 field
  spans both. Interference = Huygens superposition of the FIELD, not
  the corpuscle (Section 4.1).
- Which-path: Zone 3 field decoherence resolves the field's which-slit
  contribution at r < r_lock(T). r_lock(300K) = 340 fm; coherence
  preserved for macroscopic slit separations.
- Delayed choice: no paradox. The corpuscle's slit was always definite;
  only the still-propagating FIELD can be intercepted later (Section 5.1).
- Quantum erasure (general class): single-corpuscle erasers show
  fringes directly, unconditioned; entangled two-corpuscle erasers
  require sorting by a second measurement (Section 5.2).
- Quantum eraser (Kim et al. 2000): a distinct, two-corpuscle entangled-
  pair worked example -- no retrocausality by the same no-FTL closure
  used for entanglement generally (Section 5.3).
```

NEW PREDICTIONS (not in standard QM):

```
- Minimum electron slit width for coherent diffraction: 2*lambda_bar_e = 772 fm
  (below this, slit-edge Zone 3 fields decohere the electron winding)
- Minimum proton slit width: 2*lambda_bar_p = 0.421 fm = r_grind (nuclear hard core)
- Fringe visibility vs temperature: V(T) ~ exp(-r_sep / r_lock(T))
  where r_lock(T) = (alpha*hbar_c*r_p^2 / k_B*T)^(1/3)
```

Companion script: analysis/demos/qm_doc.py (21/21 PASS)

1. MEDIUM WAVE EQUATION AND KLEIN-GORDON

1.1 Free waves in the medium

The torsion medium's universal massless-wave speed is c = sqrt(K/rho) = 1/sqrt(epsilon_0*mu_0) (exact by SI), using K = 1/eps_0 and rho = mu_0 as established elsewhere (doc_torsion Section 3.3, doc_magnetism Section 2.1). This is the SAME speed shared by every linear, massless excitation of the medium -- both the A_g bulk-compression channel and the T_1g

transverse channel propagate at c (doc_torsion Section 3.1) -- so citing K/rho here fixes c's numeric value without claiming the field below is itself a compression/pressure oscillation. "phi" in this section is a mode-neutral placeholder: Section 1.2 gives it a mass term to become the electron/proton's own Hopf-winding matter wave (not shown to be A_g specifically, nor any other single irrep), and Section 3 separately identifies light as the T_1g mode. Neither identification is being made here -- only the shared speed c is used.

Free waves satisfy:

```
(d^2/dt^2 - c^2 * nabla^2) phi = 0
```

Dispersion: omega = c * k (massless, linear). [QM1]
(Derived from a discrete Jobson-cell lattice in the continuum limit, not an independent postulate -- analysis/quantum/lattice_dwell_time_bridge.py.)

1.2 Hopf winding rest energy -> mass gap

A Hopf winding mode (the proton) is a topological defect in the torsion medium with winding number (1,2) (p=1, q=2). The Maxwell jamming constraint (N_J = 21 cells, 3V-E = 6 exactly critical) sets the minimum size of the winding. Below this size, the winding cannot be compressed -- this creates a minimum energy:

```
E_0 = m_p * c^2  [proton rest energy = mass gap of the winding mode]
```

This generates the Compton oscillation frequency:

```
omega_C = m_p * c^2 / hbar
```

Crucially: the Compton wavelength lambda_bar = hbar / (m_p * c) = hbar_c / m_p IS the Zone 1 boundary lambda_p. The winding resonates at the Compton frequency WITHIN Zone 1 (the Maxwell critical jamming region). This is the same lambda_p that appears in r_p = 4*lambda_p (PS4) and throughout nuclear structure.

The identification lambda_bar = lambda_p is a SHARED DEFINITION, not a derived coincidence: lambda_bar = hbar_c/m_p is the same formula doc_nucleus.txt (Section 1.1) uses to define lambda_p from the outset -- it was never independently derived here and then found to coincide with the Zone 1 boundary after the fact. What IS a genuine, separate result (established in doc_nucleus.txt, not re-derived by this sentence): N_J (proton) = 21.17 ~ 21 is independently Maxwell-critical (3V-E=6), and r_p = 4*lambda_p matches CODATA to 0.02% (PS4) -- real physics content, just not what this specific identification is stating.
[QM2 PASS: omega_C = c / lambda_p confirmed to machine precision]

1.3 Klein-Gordon equation for the winding mode

With a mass gap, the dispersion relation becomes:

```
omega^2 = c^2 * k^2 + omega_C^2    [Klein-Gordon]
```

At k=0 (standing winding): omega = omega_C (pure Compton oscillation)

At k >> omega_C/c: omega -> c*k (ultra-relativistic, massless limit)

This IS the Klein-Gordon equation: (d^2/dt^2 - c^2*nabla^2)phi = -(m_p*c^2/hbar)^2*phi

[QM3 PASS: both limits verified]

(Both dispersion relations above are derived from a discrete Jobson-cell lattice model, not postulated -- lattice_dwell_time_bridge.py BR1-BR4.)

2. NON-RELATIVISTIC LIMIT: SCHRODINGER EQUATION

For non-relativistic motion (v << c, k << omega_C / c = m_p*c / hbar):

```
omega_KG = sqrt(c^2*k^2 + omega_C^2)
         = omega_C * sqrt(1 + (hbar*k / m_p*c)^2)
         ~ omega_C + hbar*k^2 / (2*m_p) + O((v/c)^4)
```

The omega_C term = rest-mass oscillation. Extract it:

```
phi(x,t) = psi(x,t) * exp(-i * m_p*c^2*t / hbar)
```

Substituting into Klein-Gordon and dropping the O((v/c)^2) correction to d^2psi/dt^2:

```
i*hbar * d(psi)/dt = -(hbar^2 / 2*m_p) * nabla^2 * psi
```

This IS the time-dependent Schrödinger equation for a free particle. [QM5 PASS]

Accuracy: The approximation is exact to order (v/c)^2. For v/c = 1/1000 (deep NR): relative error = 2.5e-7, consistent with the (v/c)^2/4 Taylor expansion.

ZERO additional assumptions were made. The Schrödinger equation follows from:

```
(a) The torsion medium (K, rho, c -- already proven)
(b) The Hopf winding mass gap E_0 = m_p*c^2 (from Maxwell jamming -- already proven)
(c) The non-relativistic approximation (standard physics -- no extra postulate)
```

Potentials (V != 0): adding an external pressure field V(x) to the medium gives:

```
i*hbar * dpsi/dt = -(hbar^2/2m)*nabla^2*psi + V(x)*psi
```

This is the full Schrödinger equation. V(x) = the local torsion medium pressure (Coulomb, gravity, nuclear) -- all derived from the same medium.

3. BORN RULE FROM WAVE ENERGY DENSITY

In the torsion medium, the energy density of a pressure wave is:

```
u(x,t) = (1/2)*rho*v^2 + (1/2)*K*strain^2 proportional to |phi(x,t)|^2
```

The probability of a detector at x absorbing the wave (detecting the particle) is proportional to the wave energy density at x:

```
P(x) = u(x) / integral u dV = |psi(x)|^2 / integral |psi|^2 dV  [Born rule]
```

This is the Born rule. It requires no additional postulate: detection = energy absorption from the medium wave, and energy density is proportional to amplitude squared.

[QM6 PASS: Born rule from medium energy density]

What this derives is the FIELD-level detection-probability density -- an ensemble/statistical claim, needing no extra postulate at that level. A separate, harder question is NOT resolved here: WHY an individual corpuscle's landing point, on any given trial, reproduces exactly this distribution (rather than some other function of the field). That question is addressed honestly, not glossed over, in Section 4.1.

(This formula uses K, the compression/A_g-type elastic constant specifically; whether a shear-type wave should instead use G, or a mode-neutral form, is not yet settled -- nor is whether a classical elastic-potential-energy formula like this applies cleanly to this medium's real micromechanics, which may be closer to a granular contact system than a continuum Hookean solid. Neither open question changes the qualitative Born-rule argument above.)

Note on wave identity: Light propagation = T_1g transverse vector wave in the torsion medium (Klein-Gordon with m=0 -> Maxwell equations; see doc_torsion Section 3.1 -- T_1g is transverse, not the v_p compression branch, which is carried by A_g). A photon is a localized T_1g wave packet (corpuscle) that IS one source of such waves;
macroscopic charge oscillations (classical EM) may generate T_1g waves through the cog/Zone 3 mechanism without individual corpuscle emission events. Whether all T_1g waves require a corpuscle source or can be generated purely classically is not yet derived from the framework. [OPEN: classical vs quantum T_1g generation -- Series 3 target. See the FRAMING NOTE above for why this mirrors a real, unresolved distinction in mainstream quantum optics -- attenuated-laser sources are Poissonian (no antibunching) and do not by themselves establish one-corpuscle-per-trial, unlike true heralded single-photon sources.] Matter (electrons, protons) = massive Hopf winding modes (Klein-Gordon with mass gap m*c^2). Light waves and matter windings are two excitation types of the same medium, not separate entities. The photon (corpuscle) is the theta=0 mode. Here theta is the winding tilt angle: it measures how far the Hopf winding departs from the massless T_1g baseline. theta=0 means no Zone 1 displacement and no rest mass; theta>0 means the winding displaces Zone 1 and acquires rest energy m*c^2. Formula: theta(m) = arcsin(8*alpha*phi*m/m_p). All matter requires theta > 0. [doc_particle_generation.txt; analysis/quantum/winding_angle.py WA1-WA2]

Pair production: when a photon's wave amplitude exceeds the threshold for nucleating a new Hopf winding (E >= 2*m_e*c^2 for e+e- pair), the medium forces a new topological winding into existence. This is the Schwinger

mechanism: field amplitude > winding nucleation threshold.
Environments where this occurs: gamma-ray burst cores (kT ~ MeV), neutron star merger ejecta, the early universe (T > 5e9 K ~ 0.4 MeV), and extreme laser fields approaching the Schwinger critical field (E_S = m_e^2*c^3/e*hbar). Note: the solar core (kT ~ 1.3 keV) is ~800x below threshold and does NOT continuously pair-produce. [See doc_nucleus Zone 2: N_J=21 for the winding nucleation condition; the photon-trapping calculation for this regime remains open]

4. DOUBLE-SLIT EXPERIMENT

4.1 Why interference occurs

The proton (or electron) is a corpuscle -- always at one definite location, never split or delocalized. Its Zone 3 field extends to infinity (falling as 1/r^3) regardless of where the corpuscle itself is. (This is the Hopf-wound, theta>0 matter-particle case; see the note after (d) below for photons, theta=0, which have no Zone 3 but follow the same mechanism.) When the corpuscle approaches a barrier with two slits:

```
(a) The CORPUSCLE goes through exactly ONE slit -- always definite, never
    "at slit A and slit B simultaneously." What spans both slits is its
    Zone 3 pressure field, which reaches both openings no matter which
    slit the corpuscle itself takes (fields fall off with distance but
    extend everywhere; they are not confined to one opening).
(b) The medium pressure field passes through BOTH slit openings
    simultaneously, whichever slit the corpuscle used.
(c) At the screen, the wave contributions from both slits
    superpose: constructive where path difference = n*lambda_dB,
    destructive otherwise. This superposition is a property of the FIELD;
    it is not a claim that the corpuscle itself was in two places.
(d) Interference pattern = standard Huygens-Fresnel diffraction for the medium
    wave. Fringe spacing = lambda_dB / d where d is slit separation.
    lambda_dB = 2*pi*hbar / (m*v) = de Broglie wavelength (from KG dispersion).
```

Photons (theta=0, no Hopf winding) have no Zone 3 shell -- a Zone 3 shell is structurally tied to having a winding at all. The field that spans both slits for a photon is instead its own T_1g wave packet's ordinary transverse spatial width (set by how the beam was collimated before the slits), not a near-field structure requiring winding. The superposition math in (c)-(d) is unchanged either way: Huygens-Fresnel superposition and the fringe-spacing formula never depend on the field being scalar (A_g) specifically, only on two coherent contributions sharing a common oscillating axis -- automatic for a scalar field, and equally true for a vector (T_1g) field whenever both slit-paths carry the same polarization (the default case for a simple, single-source setup). lambda_dB for a photon uses its momentum p = E/c directly (lambda_dB = 2*pi*hbar/p), the same relation specialized to m=0 rather than a different formula.

The corpuscle's subsequent path to the screen is drawn from the resulting interference landscape; over many trials (many corpuscles, each through one slit or the other) the fringe pattern emerges statistically. A corpuscle in transit is not directly observable -- it registers only at a nexus (a screen contact, a detector atom), where it transfers momentum to whatever it meets. This is also why a single detection event is always a single, localized click and never a smeared, continuous readout: the corpuscle itself is always at one place, and that is the only place a nexus event can occur. [analysis/quantum/double_slit_field_interference.py DS1, DS5: confirms this picture reproduces the standard fringe pattern and spacing]

Not yet derived: WHY an individual corpuscle's landing point, across many trials, is distributed as exactly |psi(x)|^2 (as opposed to some other function of the field). The field mechanics above are confirmed consistent with standard wave optics; the corpuscle-selection statistics are not independently derived from them.

4.2 Why which-path measurement destroys interference

From doc_entanglement EP3/EP6: a detector placed at one slit puts a Zone 3-active particle (detector atoms) within the phase-locking radius r_lock(T) of that slit's OWN field contribution -- not of the corpuscle's instantaneous position. Section 4.1 already establishes that the field passes through BOTH openings on every trial, regardless of which slit the corpuscle itself takes; that is what generates the interference in the first place. A detector sitting at one opening therefore couples to that opening's local field amplitude on every trial, not only on the trials where the corpuscle happens to pass through the monitored slit. This is why a which-path detector at a single slit destroys interference for the entire ensemble (matching observation), not just for the roughly half of corpuscles that actually took that slit. The Zone 3 field coupling resolves which opening's contribution the detector interacted with, which is enough to mark the path and kill the coherent cross-term for every trial.

```
r_lock(300K) = 340 fm  [Phase-lock threshold at room temperature]
```

For macroscopic slits (separation >> 340 fm), the detector itself must physically interact with the field passing through its slit to resolve the path. Simply placing a detector nearby (not in the path of either opening's own field) does NOT decohere (consistent with observation: detection requires interaction).

4.2a "Sneakily turning off" the detector does not restore interference

A crucial implication: decoherence is caused by PHYSICAL ZONE 3 COUPLING between

the particle's winding and detector atoms, NOT by whether a human reads the output.

```
Experiment: a which-path detector is "turned off" (recording stopped) but the
detector atoms remain physically in place near the beam path. The particle
pattern (two bands, no interference) persists.

Explanation:
  Detector ON and recording:   atoms within r_lock -> winding resolved -> particle pattern
  Detector "off" (not recording) but atoms present: atoms STILL within r_lock
                                -> Zone 3 coupling STILL occurs -> winding STILL resolved
                                -> particle pattern persists. "Off" means nothing.
  Detector physically removed: no atoms within r_lock -> no coupling -> interference returns.

The particle's winding does not "know" whether a human is reading the detector.
It only responds to whether another Zone 3 field is within 340 fm.
The threshold is r_lock(T) -- a physical scale, not an epistemic one.

This resolves a longstanding confusion in quantum measurement theory:
measurement = physical interaction at r < r_lock(T), not observation.
Copenhagen is vague on this distinction. Torsionverse is not.

ALSO EXPLAINS: the quantum eraser (Kim et al. 2000). Pairs whose idler photon
went through the eraser (no Zone 3 coupling forced) retain the A_g singlet ->
their sub-ensemble shows interference in coincidence counts. The "recovery" of
the pattern is not retroactive: those pairs were never decohered to begin with.
```

4.2b Generalization to the final screen (many trials, full |psi|^2 pattern)

```
Sections 4.2/4.2a describe a which-path DETECTOR at a slit; the more iconic
double-slit case is the FINAL SCREEN itself -- many trials, one localized
spot per particle, building up the full interference pattern over repeated
runs. The same mechanism applies without a new postulate: screen atoms are
Zone-3-active matter exactly like detector atoms, so a corpuscle's
traveling field resolves (couples to) whichever screen atom it first comes
within r_lock(T) of -- exactly one atom per trial, one spot per particle,
never a smeared hit, by the identical physical mechanism as Section 4.2's
slit case. Repeated over many trials, the coupling PROBABILITY at each
screen location is set by the medium wave energy density there (Section 3's
Born rule, QM6) -- so the accumulated histogram of single-particle spots
reproduces the |psi(x)|^2 interference pattern exactly, without requiring
the corpuscle itself to be smeared. This is Section 4.2's mechanism applied
at the screen instead of a slit, not a separate postulate.
```

4.2c The historical "light microscope" version (a different physical route)

```
Sections 4.2/4.2a use stationary Zone 3-active matter (detector atoms)
parked at a slit. The more commonly told version of this thought
experiment (Feynman, The Feynman Lectures on Physics, Vol. III, Chapter 1,
"Watching the Electrons") instead shines light near the slits and looks
for a scattered flash to tell which slit the particle used. This is a
DIFFERENT physical mechanism from Sections 4.2/4.2a's Zone 3 phase-lock,
not the same one restated: a photon carries no persistent Hopf winding of
its own (theta=0, Section 3), so it has no Zone 3 field to phase-lock with
anything -- it decoheres the particle by directly scattering off it
(ordinary photon-corpuscle EM scattering, the same T_1g Born-vertex
coupling of strength alpha derived in [DOC ALPHA] and used everywhere else
in this framework), not by parking a second winding nearby.

Resolving which of two slits (separation d) was used needs a photon
wavelength lambda_gamma <~ d (ordinary diffraction-limited resolution). By
p_gamma = 2*pi*hbar/lambda_gamma, such a photon transfers a recoil kick of
order 2*pi*hbar/d to the particle when it scatters. The interference
fringes are themselves separated by exactly this same momentum scale: a
particle of momentum p = 2*pi*hbar/lambda_dB spread over the angular
fringe spacing lambda_dB/d (Section 4.1) has a transverse-momentum fringe
splitting of p*(lambda_dB/d) = 2*pi*hbar/d -- identical in order of
magnitude to the recoil kick above, by construction (both trace back to
the same slit separation d). A photon able to resolve which slit therefore
ALWAYS kicks the particle by about one full fringe spacing at random,
washing out the pattern. Using longer-wavelength light (lambda_gamma >> d)
```

```
instead fails to resolve which slit at all -- and correspondingly imparts
a much smaller kick, well under the fringe scale -- so interference
survives, exactly as observed.

This is the standard uncertainty-principle argument (Delta_x ~ d implies
Delta_p >= hbar/d), not a new torsionverse postulate. What the framework
adds is the identity of the scattering agent (photon = T_1g corpuscle,
Section 3 FRAMING NOTE) and the coupling it uses (the same alpha vertex
coupling as everywhere else, not a separate ad hoc interaction) -- and, as
in 4.2a, the same underlying lesson: decoherence is a physical
interaction, not an act of observation. A scattered photon carries away
which-path information whether or not anyone ever looks at where it went.
```

4.3 New prediction: minimum slit width

Below 2*lambda_bar (reduced Compton wavelength), the slit-edge Zone 3 fields of slit atoms become comparable to the winding's Zone 3 field -> decoherence.

```
Electron: 2*lambda_bar_e = 2*hbar_c/m_e = 772 fm  [QM7]
Proton:   2*lambda_bar_p = 2*hbar_c/m_p = 0.421 fm = r_grind  [QM8]
```

Standard QM has no minimum slit width. Torsionverse predicts one, testable with sub-pm electron diffraction experiments.

5. DELAYED CHOICE AND QUANTUM ERASER

5.1 Delayed choice: paradox dissolved

Wheeler's delayed choice: the experimenter decides whether to measure which-path or allow interference only after the particle has already cleared the slit plane -- while its field is still propagating toward the screen.

In the torsionverse (same corpuscle/field split as Section 4.1, applied to timing):

```
- The CORPUSCLE went through exactly one slit, always, from the instant
  it cleared the barrier. This was never indefinite, so there is nothing
  for a later choice to retroactively fix.
- What IS still open after the corpuscle clears the slits is the FIELD:
  its Zone 3 pressure wave is still propagating toward the screen, and
  whether that propagating field gets resolved (Section 4.2's r_lock(T)
  coupling) or not is still undecided.
- "Measure now" vs. "measure later, before the field reaches the screen":
  both are choices about whether to intercept the FIELD during its
  propagation, not about the corpuscle's (already-fixed) path.
```

No retrocausality, and nothing exotic required: the corpuscle's path was already a settled fact before either choice was made. The only thing "delayed" is when a real, local, sub-c perturbation reaches the still-traveling field -- a question about signal propagation, not about rewriting the past.

[QM9 PASS: no retrocausality required]

5.2 Quantum erasure: the general class

A quantum eraser experiment first makes which-path information available (destroying interference, Section 4.2), then erases it before detection, recovering interference in a conditioned sub-ensemble. Two structurally different cases exist, both covered by mechanisms already established above -- no additional postulate is needed to distinguish them.

SINGLE-CORPUSCLE ERASERS (no entanglement): one corpuscle, one detection event. There is no separate, always-fires-regardless detector analogous to D0 below -- the corpuscle's own click IS the whole trial. Which-path marking and its erasure both act on the SAME corpuscle's Zone 3 field (Section 4.2/4.2a); fringes appear directly in the raw, unconditioned data whenever the marking is erased before detection, with no coincidence counting required. Section 4.2a's beamsplitter-eraser example is of this type.

TWO-CORPUSCLE (ENTANGLED-PAIR) ERASERS: a single initial corpuscle is converted into a correlated pair (e.g. by SPDC, Section 5.3), one member going to a position-sensitive detector (D0) that fires on EVERY trial regardless of the partner's fate, the other carrying which-path information to a separate measurement. Because D0's raw data mixes both marked and erased trials together, it never shows fringes unconditioned, with or without erasure (Section 5.3) -- recovering fringes requires sorting D0's already-recorded clicks using a SECOND, independent measurement on the partner corpuscle.

Is that second requirement a deep new principle, or just bookkeeping? Both, partly: there is no exotic physics in "two separate particles need two separate measurements" -- but the STRUCTURAL FEATURE this creates (an independent, always-firing detector whose data only reveals pattern structure once conditioned on a second measurement) is real, and is what Section 5.3 works through for one specific experiment. Whether the same A_g-singlet/G32-chain correlation mechanism used elsewhere in this framework (doc_entanglement.txt) cleanly covers the photon pairs used by most tabletop erasers is a separate, open question -- docs/open_items.txt item 17.

5.3 Kim et al. 2000: a worked two-corpuscle example

Kim et al.'s setup uses spontaneous parametric down-conversion (SPDC): just past the slits, a nonlinear crystal converts the corpuscle that crossed the double slit into an entangled PAIR -- a "signal" corpuscle continuing to a screen detector D0, and an "idler" corpuscle sent along a longer, separate

path to one of four detectors (D1-D4).

Momentum conservation in the down-conversion fixes the idler's propagation direction to depend on which slit the original corpuscle used -- ordinary conservation applied to the splitting process, not a new torsionverse postulate (see NOT YET DERIVED, below). Downstream, the idler path is either:

- kept separate (D3/D4): which slit is still recoverable from the idler alone -> the D0 clicks paired with these show NO interference.
- recombined at a beamsplitter (D1/D2): the two slit-correlated idler paths become indistinguishable before detection -> which-slit information is erased -> the D0 clicks paired with these DO show interference. This recombination is the same Huygens-superposition mechanism as Section 4.1's own double slit, just run on the idler's two paths instead of the signal's.

THE CENTRAL FACT, confirmed by the real experiment (not a torsionverse-specific claim): D0's own pattern, summed over ALL idler outcomes with no sorting, never shows interference -- with or without "erasure." Fringes appear only in the sub-ensemble of D0 clicks whose entangled idler partner later lands in D1 or D2; the D3/D4-paired sub-ensemble shows plain two-slit diffraction. This is a statement about which already-recorded clicks get grouped together, not about any click's own pattern changing in real time.

Why this needs no retrocausality (a two-corpuscle correlation, stronger than Section 5.1's single-corpuscle argument): signal and idler are correlated at the moment they are created, via an A_g-locked joint configuration (doc_entanglement.txt Section 1.2; the optical-photon case is an easy, low-threshold instance of the same mechanism, doc_entanglement.txt Section 4.3(a)). By doc_entanglement.txt's independently closed no-FTL result (Section 4.4, EQ8), resolving that correlation -- whichever corpuscle is measured first -- cannot propagate information faster than c. Sorting D0's clicks by a later idler measurement does not change D0's own already-registered data; it only reveals, after the fact, which sub-ensemble a given click belongs to.
[Consistent with quantum eraser experiments by Kim et al. 2000]

NOT YET DERIVED: the geometric mapping from "which slit" to "which idler propagation direction" is imported here from standard SPDC momentum conservation. Whether the torsion medium's own picture of down-conversion (a corpuscle splitting into two lower-energy corpuscles at a nonlinear crystal) reproduces this mapping from Jobson-cell mechanics, rather than importing it from standard nonlinear optics, is an open question -- see Section 6.

5.3.1 Creation-nexus rule: narrowing the SPDC-mapping gap (2026-10-05)

A specific hypothesis proposes a definite answer to the "NOT YET DERIVED" question just above: suppose SPDC couples to the FIELD at the crystal (which, per Section 4.1, already spans both slits there regardless of which slit the real corpuscle took) rather than to the corpuscle's already-committed slit. If the down-conversion's signal+idler phases SUM-lock to the pump (ordinary energy/momentum conservation) while their DIFFERENCE stays free to carry the slit-A-vs-slit-B information, the signal's own conditional field amplitude at D0 (given which idler path) inherits the same x-dependent relative phase an ordinary double slit would produce.

Tested in analysis/quantum/kim_eraser_creation_nexus_test.py (10/10 checks PASS): routed through the real apparatus (BSa/BSb's direct ports to D3/D4, reflected portions recombined at the shared beamsplitter BSc using ordinary 50/50 unitarity -- the same +/- sign transformation the original paper itself cites, not an added assumption), this reproduces the paper's own reported structure exactly: R01(x)=env(x)*cos^2(pi*d*x/(lambda*f)), R02(x)=env(x)*sin^2(pi*d*x/(lambda*f)), R03(x)=R04(x)=env(x)/2 (matched single-peak envelopes, no fringes) -- using only machinery already established elsewhere in this document (Section 4.1's Huygens/field mechanism, applied a second time) plus standard beamsplitter unitarity. No G32 chain, no winding, no new ontology.

What this does and does not close: the phase-sum-lock rule ITSELF remains imported from standard SPDC conservation -- this result narrows, but does not fully close, the gap above (it shows the observed pattern follows automatically once that one rule is granted; it does not derive the down-conversion process itself from Jobson-cell mechanics). Status: exploratory, under review -- see docs/open_items.txt item 16.

5.3.2 Proposed controls to isolate entanglement's role (2026-10-10)

Three controls address different confounds. C1 and C3 rule out apparatus-level confounds that are NOT specific to entanglement -- beamsplitter-count artifacts, detector-hardware/position effects -- and so are theory-agnostic: their predictions test whether the observed pattern genuinely tracks the recombination/erasure FUNCTION, not some other apparatus detail that could ALSO happen to involve entanglement but isn't limited to it. C2 is the only control that directly manipulates entanglement itself, and so is the only one that can separate "genuine entanglement" from "mere classical correlation" as

the fringe pattern's actual cause. None of the three have been built; each has a distinguishable, falsifiable prediction.

C1 -- SYMMETRIC NON-RECOMBINING CONTROL (tests whether mere apparatus complexity -- beamsplitter count -- rather than actual recombination of path information, is what produces fringes). Two ALTERNATIVE approaches to this same question, each with a different tradeoff (not complementary pieces of one combined control -- pick one):

C1a: reroute the actual erasure-bound beams (Ma's and Mb's reflected outputs) onto two separate, non-mixing beamsplitters (BSc fed by slit A alone; a new BSd fed by slit B alone -> new detectors D5, D6) instead of the shared BSc -- removing ALL erasure/recombination while keeping the same winding/path structure and beamsplitter count. TRADEOFF: this also removes the recombining phase itself -- which is part of the complexity being tested -- along with everything else, so a null result here does not cleanly isolate "beamsplitter count alone" from "no recombination at all."

C1b: achieves the same underlying test without that tradeoff -- add a second beamsplitter BSd in front of the EXISTING D4 only, relabeling its two outputs D4/D5, leaving D1/D2's real erasure process completely untouched. D4/D5 now pass through 2 beamsplitters total before detection (matching the erasure path's count) with NO merging of path information at any stage (BSd has only one input), while D1/D2 continue to show the genuine, unmodified recombination exactly as in the real experiment -- a cleaner, non-invasive isolation of the same question C1a targets.

EXPECTED outcome under ordinary optics, NOT a settled prediction: since neither variant ever coherently recombines light from both slit-paths at a single detector, textbook diffraction theory says no fringes should appear in the relevant control channel(s). But this is exactly what the control needs to establish empirically, not assume -- it is entirely possible that apparatus complexity alone (sheer beamsplitter count, imperfections in the optical elements, etc.) DOES produce fringe-like structure despite no genuine recombination, which is precisely why this test is worth running rather than dismissed as foregone. A positive result (fringes appearing anyway) would be the important, diagnostic finding -- evidence that something other than genuine erasure/recombination is responsible for the original pattern. No literature precedent found for either exact variant (search-depth check only, not exhaustive).

C2 -- REMOVE ENTANGLEMENT ITSELF (tests whether genuine entanglement, not just classical correlation, is required for D0's conditioned pattern (R01/R02) to show fringes at all -- whether fringes appear there, and how strongly, is the open question itself). Swap the BBO/SPDC crystal for a pseudo-thermal source (laser + rotating ground-glass diffuser -- standard, 60-year-old technique, Martienssen & Spiller, Am. J. Phys. 32, 919 (1964)), keeping the rest of the apparatus (double slit, PS, BSa/BSb/BSc, D0-D4) unchanged. Two candidate mechanisms, making distinguishable predictions:

(i) CLASSICAL CORRELATION / ENSEMBLE AVERAGING: predicts REDUCED BUT NONZERO fringes in R01/R02. Modeling each slit region as an independent chaotic amplitude (mirroring SPDC's own "pair from EITHER region A OR region B" exclusivity) through Kim et al.'s own published D1/D3 topology (eqs. 4-5 of the original paper) gives a hard ceiling V=1/3~=33%, for D1/D2, independent of slit count -- verified by direct Monte Carlo coincidence-counting simulation, 6/6 PASS [analysis/quantum/c2_classical_correlation_visibility_bound.py]. Real-world

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    imperfections would plausibly push an actual measurement below
    this ceiling, by an amount not derivable without choosing a
    specific implementation.
(ii) TV-SPECIFIC ENTANGLEMENT-BREAKING ENERGY RELEASE: predicts a
    CLEAN ZERO -- no entanglement means no formation/breaking energy
    budget to perturb the downstream optics. Depends on two
    prerequisites not yet resolved for photon pairs specifically:
    does the A_g-singlet/G32-chain mechanism apply to theta=0
    particles at all, and what chain-breaking physically costs.
KEY LOGICAL POINT: any nonzero fringes under C2 -- even the reduced
1/3 of hypothesis (i) -- already refute the claim that entanglement
is required for the fringe pattern to exist at all (call this Claim
A) as completely as full visibility would; only a clean-zero result
(hypothesis ii) leaves Claim A standing. A reduced-but-nonzero result
narrows the live claim to entanglement being required only for
FULL/undiminished magnitude (Claim B) -- narrower than this
document's own Section 5.3 wording above currently implies (flagged
for revision once C2 is actually run, or before any stronger claim
than Claim B is asserted here).
```

C3 -- APPARATUS-REVERSAL-EQUIVALENT (tests whether the recombining FUNCTION, not specific detector hardware/position, is what determines which data set shows fringes; distinct from a literal detector swap, which the original paper never performed -- confirmed directly from the paper's own text, not a secondary source). Physically exchange which output of BSa/BSb (transmitted vs. reflected) feeds the recombining path (toward the shared mixing beamsplitter, currently BSc, currently -> D1/D2) versus the direct path (currently -> D3/D4) -- using the same physical elements, not new ones.

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EXPECTED outcome under the recombining mechanism itself, NOT a
settled prediction (theory-agnostic -- standard QM and TV agree
identically here, since both require an actual recombination event
for interference): the fringe-carrying data set should follow the
recombining FUNCTION, not physical hardware -- whichever detectors
now perform the recombining role would show fringes, whichever now
perform the direct role would not, regardless of which original
hardware is involved. But this is exactly what the control needs to
establish empirically, not assume -- a failure of this swap (fringes
staying tied to original D1/D2 hardware/position, or appearing in
neither/both) would be the important, diagnostic finding: evidence
that something OTHER than erasure itself is responsible for the
observed pattern.
```

one actually shows, standard QM and TV interpret it identically (they do not predict different outcomes FROM EACH OTHER); their value is in testing the experimental mechanism itself empirically, not in already knowing the answer, and not in distinguishing torsionverse from standard QM. C2 is the one control that could distinguish between two live TV hypotheses, and under hypothesis (i) is already a quantitative, falsifiable classical-vs-quantum discriminator independent of torsionverse specifically. None of C1-C3 have been built. C2 hypothesis (ii)'s own grounding (does entanglement even apply to photon pairs, what does chain-breaking cost) remains open. Status: exploratory, under review.

6. STATUS SUMMARY

PROVEN (from existing medium derivations + this document):

- Schrodinger equation from Klein-Gordon NR limit [QM4-5]
- Born rule (field-level detection-probability density) from medium wave energy density [QM6; individual-event statistics: see GENUINELY OPEN below]
- Double-slit interference mechanism (Huygens in medium)
- Delayed choice: no retrocausality -- the corpuscle's slit was always definite; only the field's propagation can be intercepted [QM9]

ESSENTIALLY DERIVED (borrowed from entanglement_doc):

- Which-path decoherence at r < r_lock(T) [entanglement_doc EP3; essentially derived -- t in MeV open; QM10 check is r_lock>0 sanity only]
- Which-path via light scattering (a distinct route: ordinary photon-EM recoil, not Zone 3 phase-lock) [Section 4.2c; standard uncertainty-principle argument, no new postulate]

NEW PREDICTIONS (physical argument from Zone 3 geometry; not yet experimentally tested):

- Minimum electron slit width 2*hbar_c/m_e = 772 fm [QM7 -- script check verifies formula; labeled 'NEW PREDICTION' in script; Zone 3 argument]
- Minimum proton slit width 2*lambda_p = r_p/2 = r_grind [QM8 -- second condition IS a real check: r_p = 4*lambda_p (PS4); first condition is formula-only]

ESSENTIALLY CLOSED:

- Full Schrodinger equation with potential V(x): V = local medium pressure (Coulomb, gravity all derived from same medium; adding V is direct)
- Quantum erasure (general class, Section 5.2) and Kim et al. 2000's worked example (Section 5.3): a two-corpuscle (signal/idler) effect, distinct from Section 5.1's single-corpuscle case; D0 alone never shows interference, fringes appear only in idler-coincidence sub-ensembles; no-retrocausality grounded in doc_entanglement.txt's no-FTL closure. The A_g-singlet FORMATION grounding specifically (vs. no-FTL, which is independent of it) is under review for photon pairs -- docs/open_items.txt item 17. A specific mechanism for the interference PATTERN itself (Section 5.3.1, 2026-10-05) narrows but does not fully close the SPDC-mapping gap below -- status: exploratory, under review. Three proposed controls to isolate entanglement's own role (Section 5.3.2, 2026-10-10) are not yet built -- status: exploratory, under review.
- Pauli equation: from Dirac NR limit + E+ spinor [qm_dirac_pauli.py QD6]
- Dirac equation: from Clifford algebra + E+/E- (2I) [QD1-QD4] -- uses the standard canonical Pauli/Dirac matrices; the "T_1g generators" identification is asserted in the script's comments, not computed from an I_h representation-theory derivation. g=2 follows from the standard (sigma.v)^2 = |v|^2 Clifford identity [QD5] -- true for any Pauli-like matrix set, not specific to T_1g; proton g_p != 2 from Zone structure
- Path integral: from medium Green's function (C7 static limit = Coulomb, the one medium-specific input here). The free-particle propagator form and the H E_1=-13.6 eV result are the standard textbook Feynman kernel and Bohr formula, inserted directly rather than derived from the discrete I_h lattice (unlike QM1/QM3's lattice_dwell_time_bridge.py citation above) -- QP2-QP4 verify self-consistency of that imported form, not its origin. [QP1-QP5]

GENUINELY OPEN:

- Fringe visibility V(T) quantitative formula from r_lock(T)
- Relativistic QM with spin: Dirac + EM = QED (next paper)
- WHY an individual corpuscle's landing point, across many trials, is distributed as exactly |psi(x)|^2 (Section 4.1) -- the field-level argument above (Section 3, QM6) does not by itself establish this; a genuinely hard question, not unique to this framework (the same gap exists in de Broglie-Bohm theory's "quantum equilibrium" problem, unsolved there too after decades of work).
- Section 5.3's slit<->idler-propagation-direction mapping (SPDC momentum conservation) is imported from standard nonlinear optics, not derived from Jobson-cell medium mechanics -- whether the

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down-conversion process itself (one corpuscle splitting into two)
reproduces this from medium mechanics is not yet addressed. Section
5.3.1's creation-nexus rule (2026-10-05, exploratory) narrows this
gap (shows the OBSERVED R01-R04 structure follows automatically once
the phase-sum-lock rule is granted) but does not close it (that rule
itself is still imported, not derived).
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7. COMPANION SCRIPT

```
analysis/demos/qm_doc.py    -- 21/21 PASS  [STANDALONE -- no external deps]

Working scripts:
analysis/quantum/qm_from_medium.py    -- 10/10 PASS  [KG -> Schrodinger, Born rule]
analysis/quantum/qm_dirac_pauli.py    --  7/7  PASS  [Clifford, Dirac, Pauli, g=2]
analysis/quantum/qm_path_integral.py  --  5/5  PASS  [propagator, composition, E_1]
```

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```
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[doc_nucleus]       https://doi.org/10.5281/zenodo.22042337
```


Repository: https://doi.org/10.5281/zenodo.22105622

Code: https://github.com/Destraag/torsionverse_public